Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Agentic AI Platform · Multi-Agent Frameworks · RAG Infrastructure · HITL · LLM Evaluation · Python · Cloud-Native

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience designing and shipping agentic AI platforms, multi-agent frameworks, and RAG infrastructure end-to-end. The work I do is exactly what this role describes: architecting agentic systems where LLMs reason, plan, and orchestrate tools across complex workflows, with human-in-the-loop safety gates that ensure high-stakes decisions reach a human reviewer before any action is taken. AgentProbe is the platform layer: an open-source multi-provider LLM evaluation system with reusable agent orchestration infrastructure, tool integration patterns, retrieval-augmented generation (ChromaDB, chunking, embedding optimization), a 5-metric observability engine, and a CI-integrated regression gate - production quality from day one. Leasel is the multi-agent framework: an 8-agent LangGraph system where agents coordinate, delegate, and execute dynamic tasks across a multi-step reasoning pipeline, with conditional routing, long-term memory (SQLite knowledge repository), and deterministic guardrails that enforce human approval on every regulated decision. At Entrata, I owned production enterprise systems for 9+ years - AWS, Docker microservices, event-driven async processing, FastAPI - at 99.5% uptime with zero security incidents, delivering the reliability and scalability standards distributed AI systems require. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI & Multi-Agent Systems: Python (9yr), LangGraph, LangChain, Multi-Agent Frameworks, Agent Orchestration, Tool Invocation, Planning & Reasoning, Long-Term Memory, Contextual Reasoning, HITL, Guardrails, Anthropic API, OpenAI API, Ollama, MCP, Prompt Engineering, Claude Code, Generative AI, LLMs

RAG & LLM Evaluation: Retrieval-Augmented Generation, ChromaDB, Vector Databases, Chunking Strategies, Embeddings, Semantic Search, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Agent Evaluation, Observability, CI/CD Regression Gate, NLP, BERT, spaCy, PyTorch, scikit-learn

Production Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, FastAPI, REST APIs, RabbitMQ (event-driven async), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Linux, Distributed Systems, Performance Engineering, PHI/HIPAA Compliance, Agile/Scrum, TypeScript, React.js

PROJECTS

AgentProbe: Reusable Agentic AI Platform - Orchestration, RAG & Observability

Multi-Agent · RAG Infrastructure · LLM Evaluation · Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions CI/CD · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Reusable infrastructure platform for agent orchestration, tool integration, RAG, evaluation, and observability - purpose-built for production AI systems that need to stay reliable as models evolve. Multi-provider LLM routing (Anthropic, OpenAI, Ollama) via clean tool integration abstractions; scalable RAG platform with ChromaDB vector database, chunking optimization, and embedding selection; 5-metric observability engine (correctness, groundedness, hallucination rate, latency, token efficiency); CI-integrated regression gate that blocks quality degradation before deployment. 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) coordinating multi-step evaluation pipelines. Zero regressions across 1,240+ test cases. Open-source and enterprise-deployable.

Leasel: Multi-Agent Framework with Planning, Memory & Human-in-the-Loop

Multi-Agent · HITL · Long-Term Memory · Contextual Reasoning

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

End-to-end multi-agent framework demonstrating agentic AI across coordination, delegation, dynamic task execution, and human-in-the-loop safety - the architecture NVIDIA's AI platform team is building for enterprise workflows. **Multi-agent coordination:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with real conditional edges and dynamic task delegation based on confidence and evidence. **Long-term memory:** shared SQLite knowledge repository across all agents - documents, settings, historical incidents, release notes, and audit logs persisted across sessions. **Contextual reasoning:** agents reason across multiple evidence sources before producing ranked, citation-backed hypotheses. **HITL safety gate:** deterministic escalation to human reviewer when confidence is below threshold; guardrails enforce AI never resolves regulated decisions autonomously. 93.6% precision across 1,240 test cases.

AI on FHIR: Enterprise AI - HIPAA-Compliant System Integration

Enterprise APIs · Cloud-Native · HIPAA · Full-Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack cloud-native AI system integrating agents with enterprise data sources via REST APIs - LLM-powered NLP translates natural-language queries into structured FHIR API calls; HIPAA-compliant RBAC with authentication and full data isolation; ~60% reduction in manual retrieval time. Demonstrates the enterprise system integration pattern (integrating agents with enterprise data sources, APIs, and microservices to enable real-world actions) at compliance-grade reliability.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Agentic AI Production Deployment

- Architected and deployed a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - multi-step reasoning pipeline with RAG over regulatory documents, tool orchestration, and human review gates; translated ambiguous requirements into working enterprise automation in weeks as sole technical owner.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems design; translated complex computational research outputs for cross-functional academic stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Scalable distributed systems at production reliability** - AWS S3/EC2/RDS/Glue; Docker microservices (50% scalability improvement, 30% deployment time reduction); RabbitMQ event-driven async processing with retry logic and idempotency guarantees; Redis caching; 99.5% uptime across multi-tenant SaaS; zero security incidents across 7yr of production operation.
- **Enterprise workflow automation and data pipelines** - owned HUD compliance pipeline (federal data → 180,000 domain records) for 7 years; 200+ SQL optimizations (30% performance gain); 20+ full-stack production modules across 120+ Agile sprints; built automation pipelines eliminating 80% of manual operational workload; 100% PHI compliance.
- **Technical leadership and mentoring** - mentored 5+ engineers on system design, code quality, and production practices; 300+ peer code reviews; built internal debugging tools reducing triage time 40%; led cross-functional delivery with product, compliance, and operations stakeholders; leveraged Claude Code to accelerate engineering delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017